SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1516

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5

Code of Conduct
IS Teja Yadav(192525447) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: S Teja Yadav

Assessment Tasks
Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

- Privilege changes
- API activity
- Unauthorized access
- Suspicious cloud activity

Risk-based authentication can also be used to detect unusual behaviour.

Conclusion

The multi-cloud federation failure is primarily caused by differences in identity formats, authentication mechanisms, role definitions, and authorization policies between cloud providers.

A secure solution is to introduce **identity federation using a trusted identity provider**, standardized authentication protocols, role mapping, MFA, SSO, short-lived tokens, least-privilege authorization, and centralized monitoring.

Privacy should be protected through **data minimization, encryption, controlled attribute sharing, and secure token handling**.

Therefore, a well-designed federation architecture allows users to securely access multiple cloud providers while maintaining consistent identity, authorization, security, and privacy controls.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 Hypervisors for a Cloud Data centre Hosting Mission-Critical Applications

A **hypervisor**, also called a Virtual Machine Monitor (VMM), is software that creates and manages virtual machines (VMs) by allowing multiple operating systems to share the same physical hardware. Hypervisors are mainly classified into **Type-1** and **Type-2**. For a cloud data centre hosting mission-critical applications, the choice of hypervisor directly affects performance, security, reliability, and manageability.

Type-1 Hypervisor

A **Type-1 hypervisor**, also known as a **bare-metal hypervisor**, runs directly on the physical server hardware. There is no conventional host operating system between the hypervisor and the hardware.

Examples include **VMware ESXi, Microsoft Hyper-V, and Xen**.

The architecture can be represented as:

Physical Hardware → Type-1 Hypervisor → Virtual Machines → Guest Operating Systems → Applications

Because the hypervisor directly controls CPU, memory, storage, and network resources, it provides high performance. Hardware resources can be allocated efficiently among different VMs. This is particularly important for mission-critical applications where low latency and predictable performance are required.

Type-2 Hypervisor

A **Type-2 hypervisor**, also called a **hosted hypervisor**, runs on top of a conventional operating system. Examples include **Oracle VirtualBox and VMware Workstation**.

Its architecture is:

Physical Hardware → Host Operating System → Type-2 Hypervisor → Virtual Machines → Guest Operating Systems → Applications

Since the VM requests pass through the host operating system before reaching the hardware, additional processing overhead is introduced. This generally makes Type-2 hypervisors less suitable for large-scale production datacentres.

Performance Comparison

Type-1 hypervisors generally provide better performance because they communicate directly with the hardware. CPU scheduling, memory allocation, storage access, and network operations can be optimized by the hypervisor.

Type-2 hypervisors have an additional host OS layer. Therefore, resource access may experience additional overhead. This can result in lower performance, especially when many VMs are running simultaneously.

Security Comparison

Security is especially important for mission-critical cloud applications.

Type-1 hypervisors generally have a smaller software stack because they do not require a general-purpose host operating system. A smaller attack surface can reduce the number of vulnerabilities that attackers can exploit.

In Type-2 virtualization, both the hypervisor and the host operating system must be secured. If the host OS is compromised, attackers may potentially affect the virtualization environment and the VMs running on it.

Therefore, Type-1 hypervisors generally provide stronger isolation and are preferred in enterprise datacentres.

Manageability

Type-1 hypervisors are specifically designed for server virtualization and provide centralized management features. Administrators can create, migrate, monitor, and delete VMs and allocate resources according to workload requirements.

They also support advanced datacentre features such as:

- Live migration
- High availability
- Automated resource allocation
- VM snapshots
- Centralized monitoring

Type-2 hypervisors are easier to install and are useful for development, testing, and personal systems, but they are not normally designed for managing hundreds or thousands of productions VMs.

Justification

For a cloud datacentre hosting **mission-critical applications, Type-1 hypervisors are the better choice**. They provide higher performance, stronger isolation, better resource utilization, and centralized enterprise management.

Although Type-2 hypervisors are useful for development and testing, their dependence on a host operating system introduces additional overhead and security considerations.

Conclusion

For mission-critical cloud applications, a **Type-1 bare-metal hypervisor** should be selected because it provides better performance, security, availability, and manageability. It is therefore the standard and more appropriate architecture for enterprise and cloud datacentres.

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2. Isolation of Finance, Healthcare, and Student Workloads on the Same Hardware

Cloud providers often host workloads belonging to different organizations or departments on the same physical servers. In this scenario, finance, healthcare, and student portal applications may use the same hardware resources. Since these workloads can contain sensitive information, strong **workload isolation** is essential.

Virtualization architecture provides isolation by dividing physical resources into separate virtual machines or containers. Each VM operates as an independent computing environment with its own virtual CPU, memory, virtual disk, and virtual network interfaces.

A simplified architecture is:

Physical Server

↓

Hypervisor

↓

Finance VM | Healthcare VM | Student Portal VM

Each VM has its own operating system and applications.

Compute Isolation

The hypervisor allocates CPU resources to each VM. For example, the finance VM may receive dedicated virtual CPUs while the healthcare and student portal VMs receive their own CPU allocations.

CPU scheduling prevents one workload from directly accessing another VM's CPU state.

Resource limits and reservations can also be applied. This prevents one workload from consuming all available CPU resources and affecting other tenants.

Memory Isolation

Memory isolation is another important security feature. Each VM receives a virtual memory address space that is mapped to physical memory by the virtualization layer.

The finance VM should not be able to directly read the memory assigned to the healthcare VM. Hardware-assisted virtualization features such as memory protection and second-level address translation strengthen this isolation.

Storage Isolation

Virtual disks can be assigned separately to each VM. For example:

- Finance VM → Finance database storage
- Healthcare VM → Patient-data storage
- Student VM → Student records

Access-control mechanisms ensure that a VM cannot directly access another VM's virtual disk.

Encryption can further protect sensitive information if storage is stolen or accessed without authorization.

Network Isolation

Network virtualization allows each workload to operate in separate virtual networks or security zones.

For example:

Finance Network → Finance VM

Healthcare Network → Healthcare VM

Student Network → Student VM

Virtual switches, VLANs, virtual firewalls, security groups, and network access control policies can prevent unauthorized communication between these environments.

Two Major Attack Surfaces

1. Hypervisor Attack Surface

The hypervisor is a critical component because it controls all VMs and their access to hardware. A vulnerability in the hypervisor could potentially allow an attacker to escape from a guest VM and interact with the host or other VMs.

This type of attack is commonly referred to as a **VM escape**.

Therefore, the hypervisor must be regularly patched, securely configured, monitored, and protected using strong administrative controls.

2. Virtual Networking and Management Interfaces

The second major attack surface is the virtual networking and management layer.

Attackers may target:

- Virtual switches
- APIs
- Management consoles
- Network configuration
- Administrative interfaces
- Security-group rules

If an attacker compromises a cloud management account, they may change VM configurations, network rules, or access permissions.

Security Measures

The cloud provider should implement:

- Strong tenant isolation
- Secure hypervisor configuration
- Encryption of sensitive data
- Network segmentation
- Role-based access control
- Multi-factor authentication
- Continuous monitoring
- Regular patching
- Secure APIs
- Intrusion detection

Conclusion

Virtualization allows finance, healthcare, and student workloads to share the same physical hardware while maintaining logical separation. CPU, memory, storage, and network virtualization provide the foundation for isolation.

However, the **hypervisor and virtual management/networking layers remain major attack surfaces**. Therefore, strong access control, patching, encryption, segmentation, and continuous monitoring are required to maintain secure multi-tenant cloud environments.

3. Analysis of VM Host Utilization Snapshot and Corrective Actions

The given VM host utilization snapshot is:

Resource	Utilization/Value
CPU	92%
Memory	88%
Storage I/O latency	35 ms
Average VM boot time	170 seconds

These values indicate that the physical host is experiencing significant resource pressure. Multiple bottlenecks may be contributing to poor VM performance and long boot times.

CPU Analysis

CPU utilization is **92%**, which is very high for a virtualization host.

When CPU usage remains near maximum for long periods, the hypervisor has fewer CPU cycles available to schedule virtual CPUs. This can cause CPU contention between VMs.

If many VMs are configured with large numbers of virtual CPUs, the problem can become worse because the hypervisor must schedule more virtual CPU requests.

Possible Causes

- Too many VMs on the host
- CPU-intensive applications
- Excessive vCPU allocation
- Background processes
- Poor workload distribution

Corrective Actions

The administrator can:

1. Migrate some VMs to another host.
2. Increase the number of physical CPU cores.
3. Reduce unnecessary vCPU allocations.
4. Use load balancing.
5. Identify and optimize CPU-intensive applications.

Live migration can be used to move running VMs to another physical host with minimal service interruption.

Memory Analysis

Memory utilization is **88%**, which is also high.

When physical memory becomes insufficient, the hypervisor may use memory reclamation techniques or host-level swapping. Excessive memory pressure can significantly reduce VM performance.

Possible Causes

- Too many VMs
- Excessive memory allocation
- Memory-intensive applications
- Insufficient physical RAM
- Memory overcommitment

Corrective Actions

The administrator can:

- Add physical RAM.
- Move some VMs to another host.
- Reduce unnecessary VM memory allocations.
- Optimize applications.
- Monitor memory usage continuously.

Memory should be maintained with sufficient headroom so that temporary workload spikes do not immediately result in performance degradation.

Storage I/O Latency Analysis

The storage I/O latency is **35 ms**, which indicates relatively slow storage response for workloads requiring frequent disk access.

High storage latency can affect:

- Database operations
- VM boot operations
- Application loading
- File access

- Logging
- Virtual disk operations

When multiple VMs simultaneously perform disk operations, they may compete for the same storage resources.

Corrective Actions

The organization can:

- Upgrade to SSD/NVMe storage.
- Use faster storage arrays.
- Distribute VMs across different storage resources.
- Increase storage IOPS.
- Optimize storage caching.
- Identify VMs producing excessive I/O.

VM Boot Time Analysis

The average VM boot time is **170 seconds**, which is approximately 2 minutes and 50 seconds. This is relatively high and can be caused by a combination of CPU, memory, and storage bottlenecks.

During boot, a VM requires CPU processing, memory allocation, disk reads, operating-system initialization, and network configuration. High resource utilization can therefore directly increase boot time.

Storage latency is particularly important because the operating system must read many files from virtual disks during startup.

Overall Bottleneck

The most important problem is that **CPU and memory are already highly utilized**, while storage latency is also elevated. These conditions indicate that the host is likely **overcommitted**.

Recommended Corrective Action Plan

Immediate actions:

1. Identify the VMs consuming the highest CPU and memory.
2. Migrate non-critical VMs to other hosts.
3. Reduce unnecessary resource allocations.
4. Monitor storage I/O.

Medium-term actions:

- Implement automated workload balancing.
- Upgrade RAM.
- Add CPU capacity.
- Upgrade storage to SSD/NVMe.
- Improve VM placement policies.

Long-term action:

The cloud datacenter should implement capacity planning so that hosts are not continuously operated close to maximum utilization.

Conclusion

The snapshot indicates an **overloaded virtualization host**. CPU utilization of 92% is the most immediate compute concern, while memory utilization of 88% indicates limited memory headroom. Storage latency of 35 ms contributes to slow VM operations, including the 170-second average boot time.

The best solution is a combination of **VM migration, resource optimization, hardware upgrades, faster storage, and automated load balancing**.

4. How Software Defined Datacentre Architecture Improves Agility Compared with a Traditional Datacentre

A **Software Defined Datacentre (SDDC)** is a datacentre architecture in which computing, storage, and networking resources are virtualized and controlled largely through software.

In a traditional datacentre, physical servers, storage devices, and network equipment are usually configured manually. This makes infrastructure changes slower and more dependent on physical hardware.

The SDDC changes this model by abstracting infrastructure resources and providing centralized software-based control.

Traditional Datacentre

A traditional datacentre generally consists of:

Physical Servers → Physical Storage → Physical Network Devices

If an organization needs additional computing capacity, administrators may need to purchase and install physical servers.

Similarly, network changes may require manual configuration of switches and routers.

This process takes time and can increase operational costs.

Software Defined Datacentre

The SDDC uses virtualization technologies:

Compute Virtualization

→ Virtual Machines

Storage Virtualization

→ Virtual Storage Pools

Network Virtualization

→ Virtual Networks

These components can be centrally controlled using software and automation.

Compute Virtualization

Compute virtualization separates the operating system and application from the physical server.

A single physical server can run multiple VMs. Administrators can create or remove VMs quickly.

For example, if an application suddenly requires additional capacity, the cloud administrator can deploy additional VMs without purchasing a new physical server.

Benefits

- Faster VM deployment
- Better hardware utilization
- Dynamic resource allocation
- Workload migration

Storage Virtualization

Storage virtualization combines physical storage resources into logical storage pools.

Applications do not necessarily need to know the exact physical disk where their data is stored.

Administrators can allocate storage dynamically based on workload requirements.

For example, a database application can receive additional virtual storage without manually installing a new disk in the server.

Benefits

- Centralized storage management
- Flexible capacity allocation
- Easier backup
- Improved utilization

Network Virtualization

Network virtualization separates logical network services from physical network hardware.

Virtual switches, software-defined networking, virtual firewalls, and security groups can be created and modified through software.

For example, a new application environment can be given its own virtual network within minutes instead of manually configuring multiple physical switches.

Benefits

- Faster network configuration
- Network automation
- Better workload isolation
- Easy segmentation
- Centralized control
- Rapid scaling

Agility Improvement

The major advantage of SDDC is **agility**.

In a traditional datacentre, infrastructure changes may require:

Planning → Hardware Purchase → Installation → Configuration → Testing → Deployment

In an SDDC, the process can become:

Software Request → Automated Provisioning → Deployment

This significantly reduces deployment time.

Automation

SDDC environments can use automation and orchestration to automatically deploy VMs, allocate storage, configure networks, and enforce security policies.

This reduces manual errors and improves consistency.

Scalability

SDDC allows resources to scale according to demand.

For example, during an online examination, the number of student portal users may increase significantly. Additional compute and network resources can be allocated dynamically.

After demand decreases, resources can be released.

Conclusion

SDDC provides greater agility than traditional datacenters because it transforms physical infrastructure into programmable and software-controlled resources.

Compute virtualization enables flexible VM deployment, storage virtualization enables dynamic storage allocation, and network virtualization enables rapid network configuration.

Therefore, SDDC is better suited for modern cloud environments because it provides **automation, scalability, flexibility, centralized management, faster deployment, and efficient resource utilization**.

5. Multi-Cloud Federation Failure Due to Identity Mismatches: Secure Identity and Privacy Design

A **multi-cloud environment** uses services from multiple cloud providers. For example, an organization may use one provider for computing, another for storage, and another for specialized services.

A major challenge in multi-cloud environments is **identity management**. If different cloud providers use different identity formats, authentication systems, roles, and authorization models, identity mismatches can occur.

Identity Mismatch Problem

Suppose an organization has a user:

User: Ravi

In Cloud Provider A, Ravi may be represented as:

ravi@company.com

In Cloud Provider B, the same person may be represented using a different identifier.

Similarly, roles may differ.

For example:

Provider A:
Finance Manager

Provider B:
Finance Admin

Although these identities may represent the same individual, the cloud providers may not automatically understand that they refer to the same person.

This can cause:

- Authentication failures
- Incorrect permissions
- Duplicate accounts
- Access-control problems
- Increased administrative work
- Security risks

Federation Concept

The solution is to use **identity federation**.

Identity federation allows multiple independent systems to trust a common identity provider or establish trust relationships with each other.

Instead of creating separate passwords and accounts for every cloud provider, users authenticate with a trusted identity provider.

The basic architecture is:

User → Identity Provider → Cloud Provider A

and

User → Identity Provider → Cloud Provider B

The identity provider authenticates the user and provides trusted identity information to the cloud service.

Federated authentication technologies can use standards such as **SAML, OAuth 2.0, and OpenID Connect**, depending on the services and authentication requirements.

Secure Identity Design

A secure multi-cloud identity architecture should include a centralized identity provider.

The organization should maintain:

- Centralized identity management
- Strong authentication
- Multi-factor authentication
- Role-based access control
- Attribute-based access control where appropriate
- Short-lived credentials
- Secure token exchange
- Continuous monitoring

Single Sign-On

Single Sign-On (SSO) allows a user to authenticate once and access authorized services across multiple cloud providers.

This reduces password management complexity and improves the user experience.

However, SSO should be protected using MFA and strong session management because compromising the central identity account could affect multiple cloud services.

Role Mapping

Different providers may use different role names. Therefore, the federation system should map organizational roles to cloud-specific permissions.

For example:

Company Role: Finance Manager

↓

Cloud A: Finance-Admin

↓

Cloud B: Finance-Manager

The user receives only the permissions required for their organizational role.

This follows the **principle of least privilege**.

Privacy Protection

Multi-cloud federation must also protect user privacy.

Only the minimum identity information required by the cloud provider should be shared.

For example, a cloud service may only need:

- User identifier
- Role
- Organization
- Authentication status

It may not need unnecessary personal information.

This follows the principle of **data minimization**.

Sensitive information should be encrypted both during transmission and while stored.

Token-Based Authentication

Instead of repeatedly sending passwords to cloud providers, the identity provider can issue secure authentication tokens.

The cloud provider validates the token and grants access according to the included claims and policies.

Tokens should have:

- Short expiration times
- Secure signatures
- Appropriate scopes
- Limited permissions
- Secure transmission

This reduces the risk associated with long-lived credentials.

Trust Management

Each cloud provider must establish a trusted relationship with the organization's identity provider.

The federation architecture should define:

1. Which identity provider is trusted.
2. Which users are allowed.
3. Which attributes are shared.
4. Which roles are mapped.
5. Which services can be accessed.
6. How credentials and tokens are validated.
7. How access is revoked.

When an employee leaves the organization, their central identity should be disabled so that access to all federated cloud services is revoked.

Security Monitoring

The multi-cloud environment should maintain centralized logging and monitoring.

Security teams should monitor:

- Failed login attempts
- Unusual login locations
- Token misuse